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(b)

Suppose the response variable Y has k ordered categories

$$\text{i.e. } Y = 1, 2, \dots, k$$

$$\text{Let } \pi_j(x) = P(Y=j|x)$$

Let the cumulative probability be defined as

$$y_j(x) = P(Y \leq j|x) = \sum_{r=1}^j \pi_r(x)$$

$$\therefore \text{Category probability, } \pi_j = y_j - y_{j-1}$$

$$y_0 = 0$$

$$y_k = 1$$

Proportional Odds Model:

$$\log \left[\frac{y_j(x)}{1 - y_j(x)} \right] = \theta_j - \beta^T x \quad j \in [k-1]$$

θ_j : category specific cut point

$\Rightarrow$ same β occurs for every j

β : regression coefficient

x : covariate vector

$$\therefore y_j = F(\theta_j - \beta^T x)$$

where F is the logistic function

$$F(z) = \frac{e^z}{1 + e^z}$$

The cut pts must satisfy

$$\theta_1 < \theta_2 < \dots < \theta_{k-1}$$

so that cumulative probability is

increasing

Suppose there are n observations & n_j fall into category j

$$\Rightarrow \sum_{j=1}^k n_j = n$$

$$\Rightarrow (n_1, n_2, \dots, n_k) \sim \text{Multinomial}(n; \pi_1, \dots, \pi_k)$$

$$\therefore \text{likelihood, } L = \frac{n!}{n_1! \dots n_k!} \left[\prod_{j=1}^k \pi_j^{n_j} \right]$$

$$L \propto \prod_{j=1}^k \pi_j^{n_j}$$

Taking logarithms,

$$\begin{aligned} l &= \sum_{j=1}^k n_j \log \pi_j \\ &= \sum_{j=1}^k [n_j \log (y_j - y_{j-1})] \end{aligned}$$

$$\text{Let } R_j = \sum_{i=1}^j n_i \quad \text{and}$$

$$z_j = \frac{R_j}{n} \quad [z_k = 1]$$

Consider $\frac{y_j}{y_{j+1}}$

$$\text{with } y_j = P(Y \leq j) \text{ \& } y_{j+1} = P(Y \leq j+1)$$

$$\Rightarrow \frac{y_j}{y_{j+1}} = P(Y \leq j | Y \leq j+1) \propto \frac{z_j}{z_{j+1}}$$

$\therefore$ each stage behaves like a binomial model

$$\text{Let } \phi_j = \log \left(\frac{y_j}{y_{j+1} - y_j} \right) \quad ; \quad \frac{y_j}{y_{j+1} - y_j} = \frac{y_j / y_{j+1}}{1 - y_j / y_{j+1}}$$

$$\Rightarrow \phi_j = \text{logit} \left(\frac{y_j}{y_{j+1}} \right) \Rightarrow \frac{y_j}{y_{j+1}} = \frac{e^{\phi_j}}{1 + e^{\phi_j}} = p_j //$$

$$\text{Let } g(\phi) = \log(1 + e^\phi)$$

$$\Rightarrow g'(\phi) = \frac{e^\phi}{1 + e^\phi} = p_j$$

$$g''(\phi) = \frac{e^\phi}{(1 + e^\phi)^2} = p_j (1 - p_j)$$

$$\therefore l = n \sum_{j=1}^{k-1} [z_j \phi_j - z_{j+1} g(\phi_j)]$$

$$u_j = \frac{\partial l}{\partial \phi_j} = n [z_j - z_{j+1} g'(\phi_j)] \quad \text{--- (1)}$$

$$\Rightarrow u_j = n \left[z_j - z_{j+1} \frac{y_j}{y_{j+1}} \right]$$

For max likelihood,

$$z_j - z_{j+1} \frac{y_j}{y_{j+1}} = 0$$

$$\Rightarrow \frac{z_j}{z_{j+1}} = \frac{y_j}{y_{j+1}} = \frac{z_j}{z_{j+1}}$$

$\Rightarrow$ estimated conditional probability = observed conditional proportion

$$\frac{\partial^2 l}{\partial \phi_j^2} = -n z_{j+1} g''(\phi_j) \quad [\text{From (1)}]$$

$$I_j = -n z_{j+1} p_j (1-p_j)$$

$\therefore$ Fisher information,

$$I_j = n z_{j+1} p_j (1-p_j)$$

The model parameters are:

$$\gamma = (\theta_1, \theta_2, \dots, \theta_{k-1}, \beta_1, \dots, \beta_p)^T$$

ϕ_j depends on these parameters

$$\Rightarrow \phi_j = \phi_j(\gamma)$$

$$\Rightarrow \frac{\partial l}{\partial \gamma} = \sum_{j=1}^{k-1} \frac{\partial l}{\partial \phi_j} \frac{\partial \phi_j}{\partial \gamma}$$

Let $u(\gamma) = \frac{\partial l}{\partial \gamma}$ be score vector

and

$$A(\gamma) = -E \left[\frac{\partial^2 \ell}{\partial \gamma \partial \gamma^T} \right] \quad \text{be fisher information matrix}$$

At MLE, $u(\gamma) = 0$

$\Rightarrow$ expanding the score around current estimate $\gamma^{(t)}$,

$$u(\gamma^{(t+1)}) \approx u(\gamma^{(t)}) + \underbrace{\frac{\partial u}{\partial \gamma^T}}_{= -A(\gamma)} (\gamma^{(t+1)} - \gamma^{(t)})$$

$$\Rightarrow u^{(t)} = A^{(t)} [\gamma^{(t+1)} - \gamma^{(t)}]$$

$$\gamma^{(t+1)} = \gamma^{(t)} + [A^{(t)}]^{-1} u^{(t)}$$

This is the fisher scoring update

This is repeated until convergence. McCullagh reports that roughly 4-5 cycles were enough for 4 significant digit accuracy.